Significant of MBTI Personality Model on Decision Making in University Program Selection

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***Abstract*—**The study consists of 5 major sections which involves with the related works, methodology, result and finding, discussion and conclusion. In related works, this paper focus on Myer Briggs Type Indicator (MBTI) personality model, the issues and factors of indecisiveness and undecided students in choosing university program and the challenges of making decision which related to university environment, advising system and agents. The objectives of this study are to examine students' personality and attitudes in making decision of university program, to understand how student makes decision for the program and what the factors of his or her decision making are. The analysis process has performed and the frequencies pattern was produced by calculate the percentage of personality preferences with the factors and demographic correlates which then as a result of this paper findings.

Keywords— MBTI, Five Factor Model, personality model, advising system, undecided students, indecisive students, internal factors, external factors, expert advising system, university program.

I. INTRODUCTION

Each human being has different skills, interests and knowledge. The study of human personality has a broad and variety history in psychology with an abundance of theoretical traditions. A few personality models appeared shows a major theories including humanistic, traits, evolutionary and social learning perspective. Guion and Gottier [1] concluded that there was little evidence for the validity of personality measures in personal selection. According to Barrick and Mount [2] personality can be measures to the Five Factor Model (FFM) before examining their validity for predicting job performance in relation to a number of occupational groups and performance criteria. The criteria can be a scientific, businesses, technology and application, innovation, technical, art and other interests. Therefore, this personality model able to help students to identify their ability, type of attitudes. By using personality model, students can measure what university program is suitable for them. Students who are from high school eventually have less information about university programs and the structure. Students personally should understand what are their strengths and weaknesses because each university program has different requirements and program learning outcomes. Program learning outcomes can be access through many type of assessments or instruments such as quiz, tutorials, lab exercises, test, problem-oriented learning and examination. This is important for first step of aiming the future and student's career after graduated.

The questionnaire has been set up to understand the student's personality, strengths and weaknesses based on examination on their attitude during program selection in how and what are the factors that make their decision. There are three group of students in this condition that have major problem in making decision. First group is student who is very clear with his or her decision. Secondly, student who is indecisive and lastly, student who is undecided in decision making. This study helps to discover various factors of making decision which it can be the reasons of these three condition happens and suggests some solution for reducing student's problem in decision making. The contribution of this study is the awareness of how useful of personality model as one component in academic advising system and additional research study in the issue of drop out and university setting.

II. RELATED WORK

A. Student problems

Problem solving and decision making are concerned with the ways in which people interpret problems, form goals, search information, and combine information to arrive at solutions [3]. Many individuals could not make decision and often overlook valuable information because their lack of knowledge, personality, age and others. For example, people search very little of the available information before making their decision and in some cases they were viewing only one or two pieces of highly relevant information [4]. Students also have this kind of problems during decision making. Based on study by Ahmad.H [5], there are major factors why people difficult to make decision because decision is relatively complex and involved conscious choices and acts. He highlighted, the main problems in process of decision making are emotion, belief, risk perceptions, risk attitudes and value of information.

B. Personality Model

Five Factor Model (FFM) introduced by R.M Robert and P.J Oliver [6] has five basic dimensions: extraversion, agreeableness, conscientiousness, neuroticism and openness to experience. Based on these factors, humans can interpret their major cognitive intelligence and identify the best job or the career. Additional to that, Myer Briggs Personality Model is use in many industries and education setting to assist

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management and operation, holistic teaching and learning approach, and decision making in many ways. Marc and Eyong [7] conducted research by using personality model and relates to relationship of student undergraduate achievement and student's characteristics. P.O Terrance, E.B Leonhard et al. [8] also used personality model called Myer Briggs Type Indicator (MBTI) to do measurement of achievement in engineering study. This study stated that cognitive styles are involving both cognitive perception and processing (ordering, evaluating, structuring and others) where it appeared to be most relevant variables associated with academic achievement or mental ability, especially when such constructs are assessed by tests and other means of standardized instrumentations. In 2015, O. Kemal [9] used Five Factors Model and he found the exploratory indecisiveness is not correlated with agreeableness. Self-esteem ability was found in his 300 dataset which it was the strongest predictor of exploratory indecisiveness, follows by the factors of neuroticism and extraversion. His research predicted two subtypes of personal indecisiveness: 'exploratory' and impetuous 'indecisiveness'. The results show both exploratory and impetuous indecisiveness are significantly and positively correlated to neuroticism, and negatively correlated to other four factors. The strongest predictor of impetuous indecisiveness was found to be neuroticism, followed by self-esteem, conscientiousness, and openness to experience.

In the early days of attitude research, most investigators accepted that human behavior is guided by social attitudes. Myer Briggs Type Indicator (MBTI) is a reliable and valid instrument to measure and categorize personality and behavior. MBTI has been introduced by Katharine C. Briggs and her daughter, Isabel Briggs Myers to help people understand and know how they prefer to use their mind. Recently various papers have examined the MBTI in the workplace (Berr, Church, & Waclawski [10]; Creasy & Anantatnula, [11]; Luse, McElroy, Townsend, & DeMarie, [12]). In academic setting, K.S Debra [13] used Myer Briggs preferences and proved the statistical analysis from this model helps student academic to improve the success. There are an evidence from this study that Myers-Briggs preferences correlate to student academic difficulty in the first college in the first college year.

According to Gordon [14], undecided students are those students who "unwilling, unable, or unready to make educational and/or vocational decisions" and students who are difficulty making any decision however, may considered as indecisive. There is a clear distinction between the use of the terms 'undecided' and 'indecisive' when describing college or university student. Joe Cuseo (2005) defined that there are as many reasons for being undecided students because they have diverse interests and are excited about multiple fields of study. Additional, students indecision may simply reflect a high level motivation for learning and be active involvement in the productive process of critically evaluating and prioritizing their variety academic interests [15].

Therefore, it is important to relate the personality type with students' decision making based on demographic approach. It will reveal clear information towards student attitude and also university setting in advising or consulting students during entry registration, admission and program selection.

C. University Advising System

Advising system is a part of the education system, in which it facilitates student's academic plan, match student's skills, strengths and interests. Historically, the term academic advising denotes different uses in educational literatures for example, in some Arab universities, academic advising interest has increased due to the development of the higher educational system besides the application of the credit hour system and the new programs offered [16]. According to Pargett K.K, there are two approaches in advising students: one is called "prescriptive", where the advisor tells the student what needs to be done in order to graduate; and the other is called "developmental", where the advisor and student collaborate together to ensure that the students graduates on time. It represented in the need for interaction between the advisor and advisee, readiness and preparation for advising sessions, and the availability of right and accurate information [17]. In 2010, Barbara O. research created cases for evaluated a new academic advising model which it was splits to useful three organization delivery model: centralized, shared and decentralized. A centralized advising model is one in which an institution relies on an advising center, usually staffed by a director and professional advisors, to provide all academic advising to students. In a shared model, advising functions are split between department advisors and professional advisors. In a decentralized organizational structure, all advising services are provided by departmental faculty or staffs [18]. M.Aymar developed a prototype student advising expert system that assists the students major in Information Systems (IS) in selecting their courses for each semester towards the academic degree[19]. This object-oriented based system is use by academic advisors in their academic planning for students. Based on the methods and model, it is possible to apply it into program selection recommendation.

III. METHODOLOGY

This research finds the numbers of multiple target audience. This is deem important to avoid selection bias. Sample size of study is 270 students which involvement of five private universities in Kuala Lumpur, Malaysia. Total 120 students from University Kuala Lumpur, 19 students from Taylor University, 49 students from UCSI University, 52 students from Asia Pacific University of Technology and Innovation and lastly is 30 students from SEGI University. Most of them are the students from computing program such as Bachelor in Software Engineering, Bachelor of Mobile Computing, Bachelor of Business Information Technology, Bachelor of Mass Media; Diploma in ICT and Diploma in Software Engineering. This study takes all students from first year program. Purposive sampling places the participants in relevant

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criteria that fit the research problem. Figure 1 describes the overall process of research.

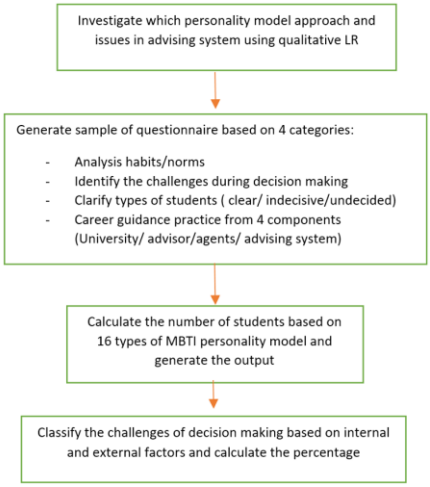

Figure 1 : Overall Research Process

A. Questionaire and Data Analysis

The data collection has been analyzed by four designed categories of independent variables : category A is examination on student's norms and behaviors with the objective is to identify user's personality; category B is identify the challenges during decision making with objective is to understand the problem and situation; category C is identification on indecisiveness and undecided type of students with the objective is to identify either user is undecided or indecisive person; category D is examination on what are common practices of university advising system and university support with two objectives which that to investigate how an advising system supports students in high school and during admission process, and to observe interpretation of students' on academic advising system.

Research uses 7 Liked-scale format to avoid contradict findings based on user dilemma as example, Joshi et al.(2015) stated that the dilemma of choice and explicit greater extent of measurement by 7 point scale is very much in the territory of the reason of response without which consideration of reliability is of number weight.

It may create the 'curves of reliability' around the 'zenith of validity'. Based on previous study, research assumed the output for all categories as in Figure 2.

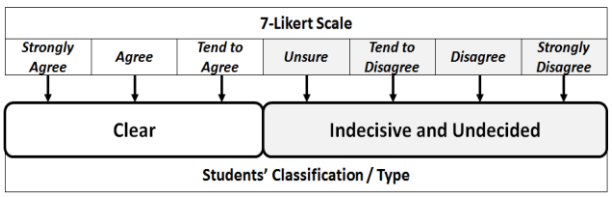

Figure 2 : Classification Of Indecisive And Undecided Student Based On Liked-Scale Formal

B. MBTI Personality Model

MBTI model has 16 type of personality based on unique combinations of the preferences as shows in Figure 3. The combinations are formed from the four basic types: (1)introvert or extrovert, (2)sensing or intuation, (3)thinking or feeling and (4)perceiving or judging. Research studied each of the possible types and matched it with every selected program learning outcomes. Four letters are used to represent a type, for example a person with preferences for **E**xtraverted, **S**ensing, **T**hinking, **J**udging is called an ESTJ.

ST	SF	NF	NT
Business/Management Law/Law Enforcement Skilled Trades	Health Care Teaching/Education	Counseling Arts Teaching/Education Religion	Science Technology Management

ISTJ **Occupational Trends:** Management Administration Law Enforcement Accounting Focus: use attention to detail to get the job done.	**ISFJ** **Occupational Trends:** Education Health Care Religious Settings Focus: personally help people behind the scenes	**INFJ** **Occupational Trends:** Religion Counseling Teaching Arts Focus: facilitate others' emotional, intellectual or spiritual development	**INTJ** **Occupational Trends:** Scientific/Technical Field Computers Law Focus: use intellectual creativity and technical knowledge to conceptualize, analyze and get tasks done
ISTP **Occupational Trends:** Skilled Trades Agriculture Law Enforcement Military Technical Fields Focus: work hands-on with data or things and analyze	**ISFP** **Occupational Trends:** Health Care Business Law Enforcement Focus: use gentle nature and attention to detail, in service-related positions	**INFP** **Occupational Trends:** Counseling Writing Arts Focus: use creativity to honor or support their values	**INTP** **Occupational Trends:** Scientific or Technical Fields Focus: apply objective analysis of problems and technical expertise in solitary environment
ESTP **Occupational Trends:** Marketing Skilled Trades Business Law Enforcement Focus: take action, work hands-on and attend to details	**ESFP** **Occupational Trends:** Health Care Teaching Child Care Skilled Trades Focus: use outgoing nature and enthusiasm to help people with practical needs	**ENFP** **Occupational Trends:** Counseling Teaching Religion Arts Focus: use creativity and communication to foster growth in others	**ENTP** **Occupational Trends:** Science Management Technology Arts Focus: continuously take on new challenges and solve problems
ESTJ **Occupational Trends:** Management Administration Law Enforcement Focus: use logic and organization of the facts to get tasks done	**ESFJ** **Occupational Trends:** Education Health Care Religion Focus: use their personal concern to provide service to others	**ENFJ** **Occupational Trends:** Religion Arts Teaching Focus: help others with their emotional, intellectual and spiritual growth	**ENTJ** **Occupational Trends:** Management Leadership Focus: use analysis, strategic planning, and organization to get tasks completed

Figure 3 : MBTI Manual And Qualifying Program

All questions in Category A is based on the MBTI manual which considered logical explorer of habits and norms. A few keywords or indicators have been used to generate questions and mapped to these 16 types of preferences for example the skills, logical thinking, focus action, responds in certain situation, learning process and others. The score calculated by giving 1 to each selected preferences by students. For example, in category A: analysis on habits or norms has a question of "*I usually plan ahead for my day.*" If student answer 'strongly agree' or 'agree' or 'tend to agree', it means this student has 'judging' preference and this preference gets 1 score and automatically 'perceiving' preference gets 0 score. Research also required gender and type of program into this analysis for future record.

C. The Challenges of Decision Making

Research already set the internal and external factors that influenced the direction of decision making as shows in Table 2 which can be found in Category B. Each question in category B has a few key words related to each of factors. If

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student answer 'strongly agree' or 'agree' or 'tend to agree', the score is 1, if the answer is 'unsure' the score is 0.5 and if the answer is "strongly disagree' or 'disagree' or 'tend to disagree', the score is 0. For example in category B has a question of "*My family is supportive of the current program that I chose*". This question triggers 'family' as one of external factors. If student answer such as 'strongly agree", therefore 1 score gives to external factors under 'family' variable as shows in Table 1. However, there are other questions in different category, (extraction from Category C) which research added the weightage in the same variable;family. For example in category C has a question of "*I choose the program based on family recommendation*". The score from this category also considered as an external factor in category B because family is one of variables in external factors.

Table 1 : Score Based On Challenges Setting From Category B

Family				Remark	Financial		Remark	Peer		Remark	University						Remark
B1	B2	B4	B16		B12	B16		B3	B5		B5	B6	B10	B13	B14	B15	
1	0	0	1	0.5	0.5	1	0	0	1	0.5	1	1	1	0.5	1	0	1
1	1	1	1	1	1	1	0	1	0.5	1	0.5	1	0	1	1	1	1
1	0	0	1	0.5	1	1	0	1	1	1	1	1	1	1	1	0	1
1	1	1	1	1	0.5	1	0	1	1	1	1	1	1	1	1	0.5	1
1	0	0	0	0	1	0	0.5	0	1	0.5	1	0.5	1	0	0.5	1	1
1	0	0	1	0.5	0.5	1	0	0	1	0.5	1	0	1	0	0	1	0.5
1	0.5	1	0.5	1	1	0.5	0	1	1	1	1	0.5	1	1	1	0	1
1	0	0	1	0.5	1	1	0	0	1	0.5	1	1	1	0	0.5	1	1

IV RESULT AND FINDINGS

Research has three domain components of result. Firstly, research found that 'undecided and indecisive' students is existed. In Table 2, to identify either the student is 'clear' make his or her decision based on internal factor, the result took from seven selected questions (C1,C2,C8,C9,C10,C11,C12) in Category C and for external factors, the result took from question(C3-university variable),(C4-advising system variable),(C5- family variable), question (C6- friend or peer variable) and question (C7- agent variable). Result shows only (18.2%) students are clear or made their own decision, indecisive students (52%) and undecided students (30.5%). It shows clear students influenced by external factors such as university (22.4%) and family(17%). Students said that they understood the university program structure, university provided proper information and thus, they able to determine their path way. Secondly, research generates a cumulative result for factors or challenges that influenced undecided and indecisive of choice made. The result shows in external factor, 65 of friends or peer and 63% of agents are two major influence variables that drive to student cognitively in decision making. Therefore, the external factor such as university support, program information and course structure of program are significant on shaping students' self-confident besides involvement of family. Overall result, internal factor developed the highest factor than external factor with percentage 30% to 54%. It becomes the biggest obstacle due to difficulty to do inductive reasoning, conclusion and influence by friends and agent. They cannot identified their self-esteem, unsure with certain degree of information of what they were belief and other issues related to the personality.

Thirdly, research found a significant of using personality in identify internal factor of decision making. Each universities has the requirements and academic's aims. For example, Bachelor in Software Engineering, the learning outcomes are student has practical hands on activity in lab, assessment is a group assignment including development of critical thinking in class and exam. Research studied which MBTI preference is suitable to determine what type of students can take the computer programs.

Table 2 : Overall Result Of Indecisive And Undecided Students Together With Factors Of Decision Making

No.	Factor	Clear (respondent)	%	Indecisive (respondent)	%	Undecided (respondent)	%
1.	**Internal** (Habit, norm or personality)	71	18.20	201	51.53	118	30.25
	External	**Clear**		**Indecisive**		**Undecided**	
2.	University (Course & Program)	66	22.44	146	49.66	82	27.89
3.	Advising (Advisor & Advising system)	19	6.09	139	44.55	154	49.36
4.	Family	55	17.18	129	40.31	136	42.50
5.	Friend's or peer	7	2.22	103	32.70	205	65.08
6.	Agent	9	2.84	107	33.75	201	63.40

Table 3 shows 88 respondents were ESFJ and secondly, 33 respondents were ISTJ. Meanwhile, ESFJ type required student who is good in using their personal concern to provide service to others. The best program recommended is Multimedia Technology (MMT) and Game Development (GD). Besides, student has ISTJ type needs to be ready and get job done on time. It required student who is good in analysis and particular to the details and program recommended such as Software Engineering(SE), Computer Science(CS), Intelligence System (IS), Information Security(ISe) and Information Technology (IT).

Table 3 : Number Of Students Under Difference Program In Computer Study Based On MBTI Preferences

Category of MBTI	Number of Students
ISTJ	33
INFJ	18
INTJ	6
ISTP	8
ISFP	11
ISFJ	20
INFP	10
INTP	7
ESTP	6
ESFP	22
ENTP	8
ENFP	20
ESTJ	14
ESFJ	88
ENTJ	9
ENFJ	20

Based on MBTI manual, research analyzed which features of personality has reflected to the program in university. For example, 'BSc in Software Engineering' required student who is practical, good in technical development, hands on with data and logical assimilators. This program suitable for student that has ISTP, ENTP, ESTP, ESTJ and ISTJ. Additional, 'BSc in Mobile Technology (MT)' required student who is interested in knowledge of design, development and fast-moving technology. This program is suitable for student that has INTP, ISTP, ENTP,

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ENTJ and ESTJ type. If student is more features of ESTJ type, it means he or she has greater chance to achieve highest score in MT program. Finally, there are 22 students are in the category of ESFP type. Student from this preference must be compassionate responders which use outgoing nature and enthusiasm to help other people with practical needs. Program such as Information Technology (IT) is suitable for them.

V. DISCUSSION

Based on this research, "indecisive" and "undecided" group of students is existed in academic program decision making. External factor is a major challenge to these group of students. Research found the variables such as university environment, information, course of program structure and family from external factors become major variables influenced 'clear' students in decision making. These variables are the highest in overall factors. Second type of variable is a factor of peer with 74 respondents as shows in Figure 4.

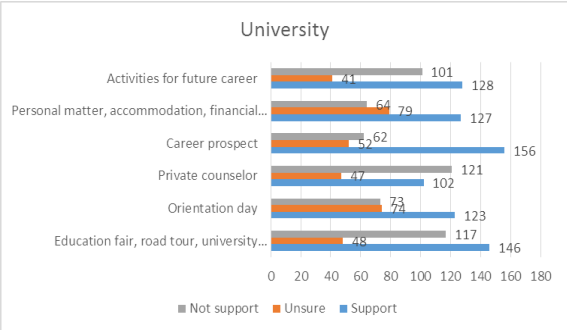

Figure 4 : The Variables Involved In University Environment, Information And Advising System

University information, the attitude of advisors, communication between student and university staffs and job prospect based on program are the most important indicators to ensure student motivation and clear in making decision. University activities such as tour in university, education fair, orientation day, open day and promotion in school tour helped student gets more value of information and able to be self-select person and on the right track of program. Besides, university advising system needs to improve its system in term of the fast response, process of registration, personality identification and in a way of recommendation. University should implement expert advising system such as Artificial Neural Fuzzy Inference System to help student in recognized his or her personality and the possible path way.

Other than that, the advisor must perform well communication in explaining program structure and career prospect. They must able to identify the strengths and weaknesses of students based on selected program and also considered what other expert's advice. This can be done by using an automated expert advising system that including personality test. Additional, student should recognized their personality before they choose what program they should takes. By using personality test, student can identify skills and possible target career. Despite the qualification is needed as one of admission requirements, students must understand their ability and skills (internal factors), either they should or should not enroll the program. Each program has different career prospect and different level of difficulties.

In this research, there are seven programs related to MBTI preferences as shows in Table 4. With the presentation of results and findings in this paper, it is significant to state that the root causes of an undergraduate student's future should start from right on the time of they decided on the academic program during admission and it is must to consider all rational internal and external factors in reducing the number of drop out in semesters or student intends to change his or her programs during study.

Table 4 : Program Mapped To Mbti Personality Model

Program	MBTI Preference
Software Engineering	INTJ, INTP, ISTP, ISFP, ISFJ, ISTJ, ENTP, ENTJ, ENFP, ESTP, ESFP
Information Technology	INTJ, INTP, INFJ, ISTP, ISFP, ISFJ, ESTP, ESFP
Multimedia Technology	INFP, ESFJ
Games Development	INFP, ESFJ
Mobile Technology	ENTJ, ESTJ
Information Security	ISTP, ISTJ, ENTP, ENTJ, ESTJ
Intelligent System	ISTP, ISTJ, ENTJ, ENFP

VI. CONCLUSION

By obtained the result from this dataset, all three objectives are achieved. Firstly, personality model is important as a tool in advising system because students rely on their belief, emotion, valuable information based on external factors and human's nature. Personality model can be used in analyzes students ability and skills in university advising system not only during semester, it also can be used at the beginning of admission process to give a few recommendation of what program that student suitable with instead by only depending on student's preferences, qualification and university requirements. MBTI is selected in this case because it has sixteen types of personality, it has details evaluation that being using more than decades and proven. Secondly, this study proves that there are many students are undecided and indecisive. Lastly, the factors that influenced their decision making are based on variables as shows in Table 2. The external factors such as university environment, university information, advisor and university activities, friends and peer are important to reduce or avoid the group of indecisive and undecided. Therefore, MBTI personality model is recommended to support advising system and decision making during university program selection.

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